

Analytics Engineer Challenge

Air Cote d'Ivoire growth allocation challenge — candidate brief

Purpose

Use this pack to send a candidate challenge, score submissions consistently, and calibrate interviewers across junior, mid-level, and senior Analytics Engineer profiles.

Business context

Air Cote d'Ivoire is the flag carrier of Cote d'Ivoire, operating domestic, regional, and international routes from Abidjan. Its recent fleet expansion with Airbus A330-900 aircraft supports long-haul ambitions, starting with Paris and potentially broader expansion toward the Middle East and North America. The airline also operates adjacent products such as loyalty and cargo, making growth decisions inherently cross-functional across network, operations, and customer teams.

Your mission is to build an analytics product that helps leadership decide where to allocate constrained budget over the next 12 months to maximize profitable growth.

Decision question

Where should Air Cote d'Ivoire invest first to maximize profitable growth: route expansion and optimization, customer retention, or upsell / cross-sell?

- Identify growth opportunities by route, market, or customer segment.
- Recommend better budget allocation across network growth, retention, and ancillary revenue.
- Show how both structured and unstructured data can improve decision quality.
- Expose the resulting analytics through a dashboard and a lightweight AI interface.

Starter data and enrichment freedom

You will receive a starter dataset with: Flights, Bookings, Customers, Airports, and Routes.

- You are encouraged to enrich the dataset by creating synthetic data and additional entities when they improve the analysis.
- Examples include Aircraft, manufacturer, fleet assignment, fuel cost, fare family, ancillaries, loyalty activity, support tickets, disruption logs, baggage events, weather proxies, crew, or competitor context.
- Explain which new data you created, why it matters, and how it changes the business recommendations.

What we want to evaluate

Topic	Guidance
Context and customer need understanding	Can you frame the airline problem correctly, identify the decision makers, and generate useful data rather than random data?
Data processing and modeling	Can you create a model that supports dashboards, decision-making, and AI use cases across structured and unstructured sources?
Data visualization	Can you turn the model into a decision-oriented dashboard instead of a collection of charts?
Agentic AI	Can you expose the data through a simple MCP server or equivalent tool layer so an AI assistant can reason over it?

Challenge tasks

Part 1 — Business understanding and data generation

- Describe the airline business domains that matter for the decision question: network, operations, commercial, customer, and ancillary revenue.
- List the KPIs that should guide decisions. Examples: route revenue, route margin, load factor, delay rate, cancellation rate, repeat booking rate, ancillary attach rate, loyalty engagement, customer sentiment.
- Generate or enrich realistic synthetic data, including at least one unstructured dataset such as reviews, support tickets, complaint logs, or operational notes.
- Document your assumptions and explain why your added data is useful.

Part 2 — Data processing, modeling, semantic layer, and ontology

- Build a model that supports the three business themes: route optimization and growth, customer retention, and upsell / cross-sell.
- You may use a star schema, Data Vault, or a hybrid design. Justify the choice.
- Create a business-friendly semantic layer that defines core entities, KPI definitions, joins, and naming conventions.
- For senior differentiation, add an ontology-inspired layer or explicit reasoning rules that classify business concepts such as High-Value At-Risk Customer or Strategic but Underperforming Route.
- Show how unstructured data is integrated. Examples: sentiment scoring, complaint categories, route-level issue themes, or semantic tags.

Concept notes for candidates:

Concept	Explanation
Star schema	A dimensional model optimized for analytics, typically with fact tables for events and dimension tables for descriptive context.

Data Vault	A modeling approach designed for scalability, historization, and auditability using hubs, links, and satellites.
Semantic layer	A business-friendly layer that maps business terms such as revenue or load factor to the underlying tables and calculations.
Ontology	A formal representation of concepts, relationships, and rules that can support validation and inference beyond simple query translation.
MCP	Model Context Protocol, an open protocol that lets AI assistants use tools and external data sources in a structured way.

Part 3 — Dashboard

- Build an Executive Growth Allocation Dashboard using the BI or app tool of your choice.
- It should cover: network and profitability; customer and retention; upsell and cross-sell; and a final recommendation layer.
- Provide screenshots, a live link, or a short video walkthrough.
- Include one page of executive recommendations explaining what the airline should do next and why.

Dashboard area	Minimum expectations
Network and profitability	Revenue, margin, yield, load factor, delays, cancellations, demand trends, and route opportunity matrix.
Customer and retention	Segmentation, repeat behavior, loyalty, high-value customers at risk, complaint themes, and satisfaction trends.
Upsell and cross-sell	Upgrade conversion, ancillary attach rate, offer propensity, and revenue per passenger or segment.
Decision layer	Routes to grow, routes to defend, customers to retain, and offers to prioritize.

Part 4 — Agentic AI

- Expose your modeled data through a small MCP server or an equivalent tool layer that an AI assistant can use.
- The AI interface must be able to use both structured data and at least one unstructured source.
- Demonstrate a few grounded questions, such as: Which routes deserve more budget next quarter? Which high-value customers are at risk? What complaints are driving low satisfaction on route X?
- Provide a video showing the interaction and explain the architecture briefly.

Level expectations

Level	What stronger submissions typically include
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Junior	Clear business framing, realistic synthetic data, a sound dimensional model, a useful dashboard, and a simple AI interaction.
Mid-level	Richer transformations, stronger metric definitions, better unstructured enrichment, and more actionable recommendations.
Senior	Deliberate modeling trade-offs, a reusable semantic layer, ontology-inspired inference, robust AI tooling, and strong documentation.

Deliverables

- Source code repository or folder containing the pipeline, models, dashboard assets, and AI interface.
- README with setup and run instructions.
- Data dictionary and modeling diagram.
- Screenshots, links, or short video for the dashboard.
- Short demo video for the AI interaction.
- A short write-up covering assumptions, architecture, limitations, and next steps.

Suggested acceptance questions

- Which routes should receive more budget next quarter, and why?
- Which routes look unprofitable because of operational issues rather than weak demand?
- Which customers are high value and at risk of churn?
- Which customer segments should receive premium upgrade or ancillary offers?
- Compare two routes using both financial and satisfaction signals.
- Explain the evidence behind one recommendation using both structured and unstructured data.

Recommended documentation

[dbt Semantic Layer](#)

[dbt Semantic Models](#)

[Model Context Protocol introduction](#)

[MCP specification](#)

[W3C RDF 1.1 Primer](#)

[OWL 2 Primer](#)

[SHACL Recommendation](#)

[Apache Superset documentation](#)

[DuckDB + dbt setup guide](#)

Reference context used to shape this challenge

Official and reference material consulted:

[Air Cote d'Ivoire official site](#)

[Air Cote d'Ivoire A330neo announcement](#)

[Deep-dive article: Semantic Layers Translate, Ontologies Reason](#)

[OpenMetadata video reference](#)